Assignment 1 - Reinforcement Learning with Applications to Finance

Finite-Horizon Asset Allocation with CARA Utility

RONG Zhixia | Student ID: 21269362

Course deliverable: model formulation, numerical solution, and empirical validation

Executive Summary

This report formulates the portfolio allocation problem as a finite-horizon Markov Decision Process (MDP). The investor allocates wealth across cash and risky assets, faces a 10% turnover constraint each period, and evaluates outcomes with a Constant Absolute Risk Aversion (CARA) utility function. A discretized state-action space and sampled backward induction are used to obtain the optimal policy. The main findings are:

1. In the **n = 3** cases, the learned policy outperforms buy-and-hold in both average terminal wealth and expected utility.
2. The **optimal initial action changes with the horizon**, which is consistent with a finite-horizon control problem.
3. In the **n = 4, T = 5** case, average terminal wealth increases slightly, but expected utility does not, showing that a higher mean payoff does not always align with the preferences of a risk-averse investor.
4. Computational cost rises sharply as the number of assets and the horizon increase, which highlights an important practical limitation of the tabular discretization approach.

At a glance	Initial action	Avg wealth lift	Utility result
n = 3, T = 5	(0.3, 0.2, 0.3, 0.2)	+0.0611 vs B&H	Improved
n = 3, T = 8	(0.3, 0.2, 0.2, 0.3)	+0.0649 vs B&H	Improved
n = 4, T = 5	(0.3, 0.0, 0.3, 0.2, 0.2)	+0.0165 vs B&H	Not improved

1. Problem Formulation and Modeling Logic

The assignment considers a discrete-time portfolio problem with cash and risky assets. At each date, the investor may rebalance at most 10% of the portfolio and seeks to maximize terminal utility. To turn the problem into a sequential decision model, the portfolio problem is formulated as a finite-horizon MDP.

MDP components

- **State.** The state records time, current wealth, and current portfolio weights.
- **Action.** The action is the post-trade target weight vector subject to feasibility and the turnover cap.
- **Transition.** Wealth evolves with the realized risky returns and the cash return; the next-period weights drift endogenously after returns are realized.
- **Reward.** Intermediate reward is zero and terminal reward is CARA utility.

$$s_t = (t,\ W_t,\ p_t)$$

State definition

$$\frac{1}{2}\sum_{k=0}^{n} |q_{t,k} - p_{t,k}| \leq 0.10$$

Turnover constraint

$$W_{t+1} = W_t\left[q_{t,0}(1+r) + \sum_{k=1}^{n} q_{t,k}(1+R_{t+1,k})\right]$$

Wealth transition

$$U(w) = -e^{-\alpha w}, \qquad \max_{\pi} \mathbb{E}^{\pi}[U(W_T)]$$

Reward and objective

2. Numerical Method

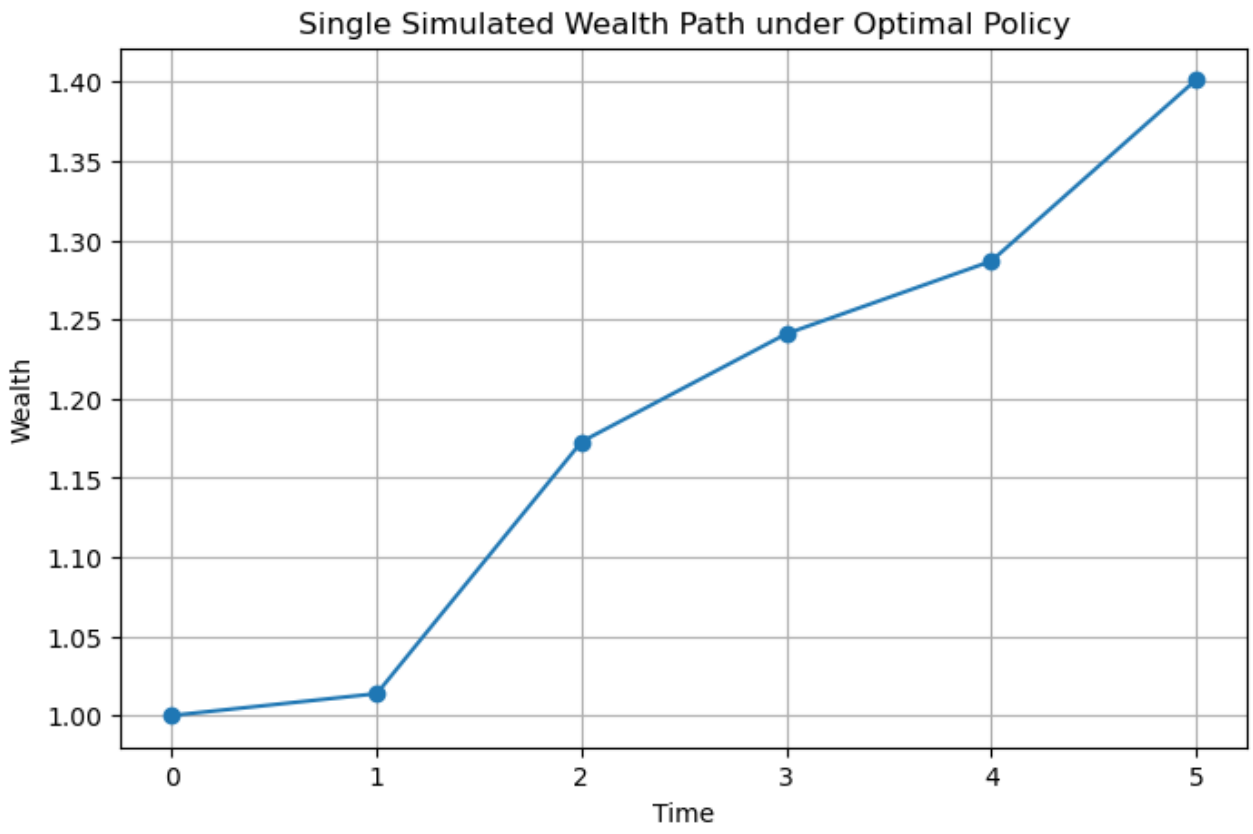
The original problem has continuous states and continuous actions. To make the problem computationally tractable, wealth is discretized into bins and portfolio weights are placed on a grid. Given the assignment scale, sampled backward induction is used: for each state-action pair, Monte Carlo sampling approximates the continuation value, and the action with the largest expected value is selected.

This design is simple and transparent, but it is computationally expensive. The runtime grows rapidly with the number of risky assets, the horizon, and the fineness of the discretization grid.

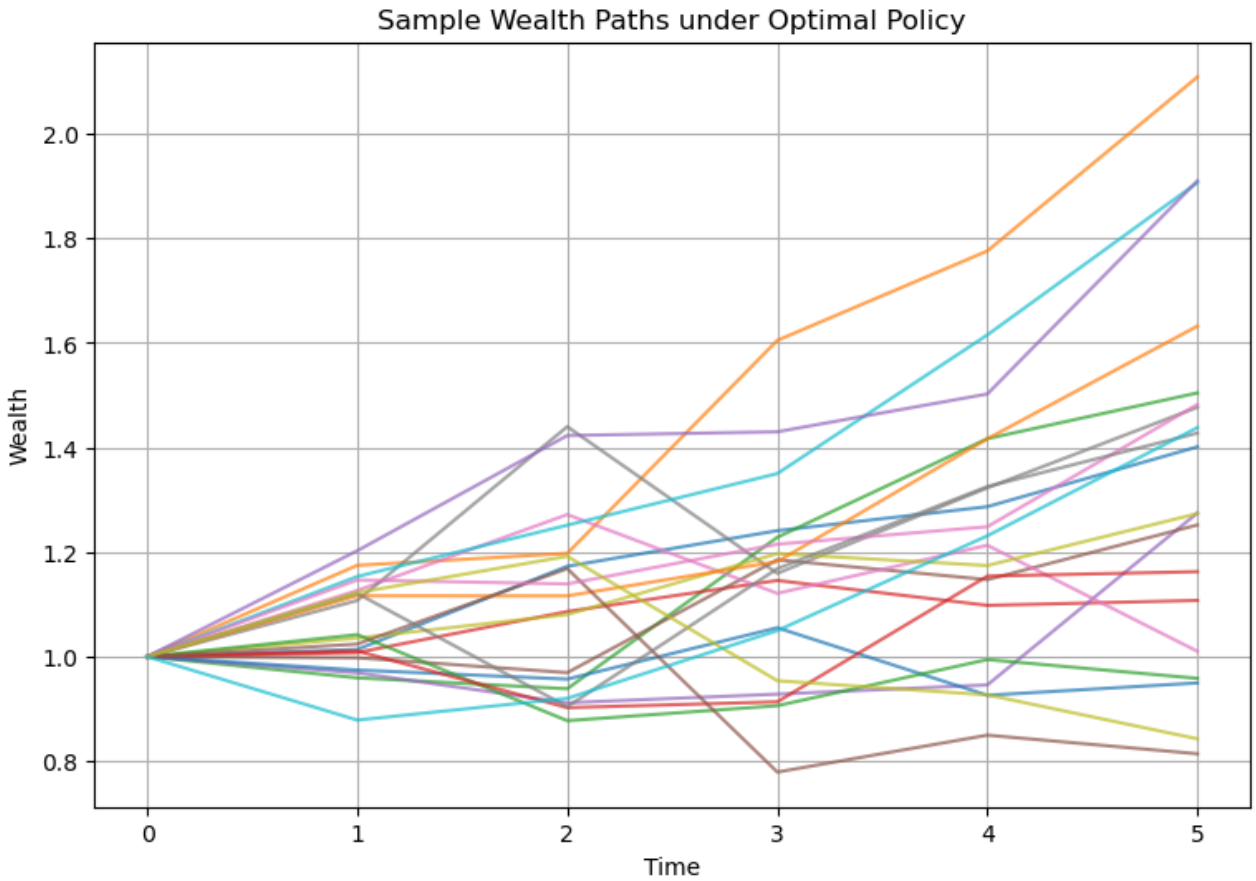
3. Main Numerical Results: n = 3, T = 5

The main benchmark case uses three risky assets and a 5-period horizon. The learned initial action is (0.3, 0.2, 0.3, 0.2), which shifts 10% of wealth from cash to the second risky asset. This is economically intuitive because the second risky asset has the highest expected return in the chosen parameter setting.

Metric	Optimal Policy	Buy-and-Hold
Average terminal wealth	1.3010	1.2399
Std terminal wealth	0.3532	0.2372
Average terminal utility	-0.2884	-0.2972
Min terminal wealth	0.6795	0.8171
Max terminal wealth	2.4602	1.9386

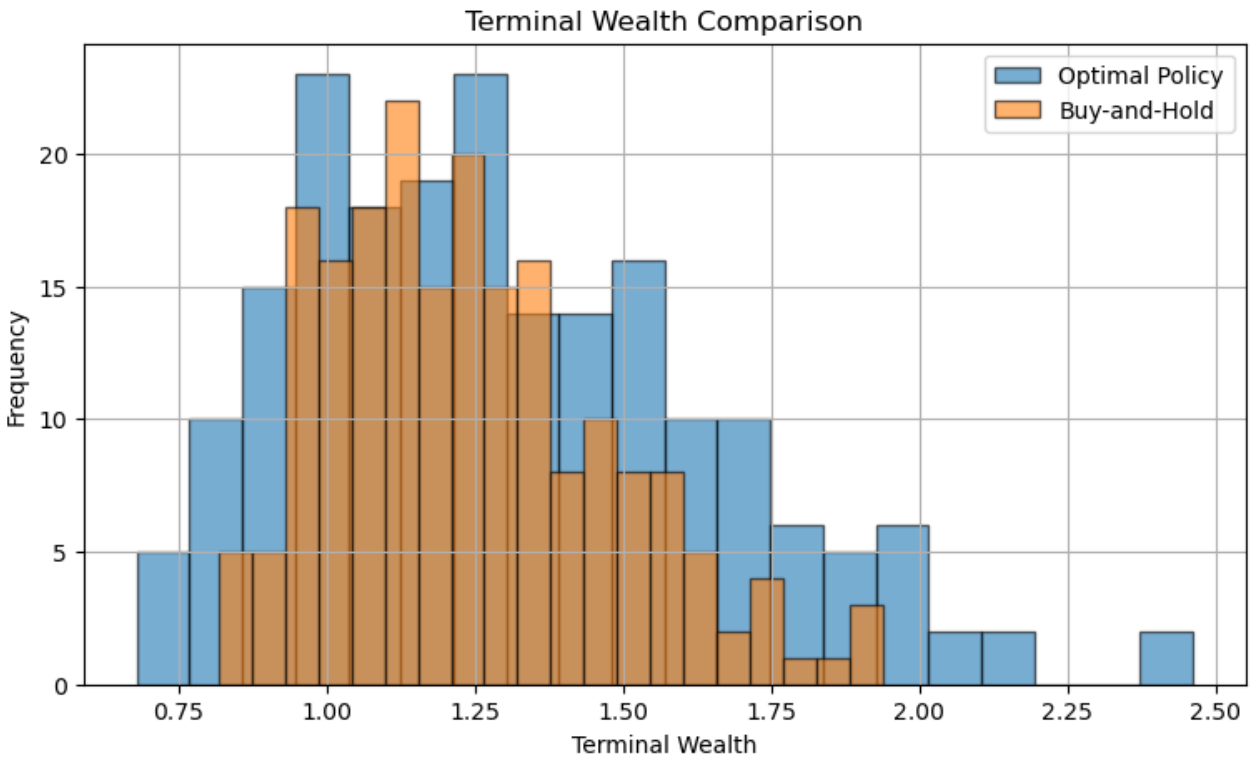


Single-path wealth trajectory under the optimal policy.



A sample of simulated wealth paths under the learned policy.

4. Comparison and Interpretation



Terminal wealth distribution: the optimal policy shifts mass to the right and also produces a wider dispersion.

In the main case, the optimal policy dominates buy-and-hold in both mean terminal wealth and expected utility. At the same time, the standard deviation and the left tail are somewhat worse than in buy-and-hold, which is visible in the histogram and in the minimum terminal wealth. Therefore, the learned strategy improves average performance by taking more risk, but in this case the trade-off is still favorable for a CARA investor.

Horizon effect and higher-dimensional case

When the horizon increases from T = 5 to T = 8 with n = 3, the initial action changes from (0.3, 0.2, 0.3, 0.2) to (0.3, 0.2, 0.2, 0.3). This confirms that the policy is genuinely finite-horizon: the best action depends on the remaining time to maturity.

For n = 4 and T = 5, the average terminal wealth of the optimal policy remains slightly above buy-and-hold, but the expected utility becomes worse. This is an important result rather than a failure: maximizing the mean payoff and maximizing expected utility are not always aligned, especially for a risk-averse investor.

5. Conclusions and Practical Limitation

This assignment shows that a finite-horizon portfolio problem can be expressed cleanly as an MDP and solved approximately using discretization plus sampled backward induction. The numerical results demonstrate that the approach can produce economically meaningful policies. In the n = 3 experiments, the learned strategy improves both average terminal wealth and expected utility relative to buy-and-hold.

The experiments also reveal two important lessons. First, the optimal allocation depends on the remaining horizon, which is consistent with the theory of finite-horizon dynamic control. Second, better average wealth does not automatically imply better utility: the n = 4, T = 5 case shows that a policy can raise the mean payoff while becoming less attractive to a CARA investor.

Computational note

The case n = 4, T = 8 was initiated but not fully included in the main discussion because the state-action space becomes very large under a 0.1 weight grid. This illustrates the main weakness of the tabular discretization approach: although it is easy to interpret, the computational burden rises sharply as the number of assets and the horizon expand.

Key takeaways

- The MDP formulation is appropriate for dynamic portfolio choice with trading constraints.
- The learned policy is horizon-dependent and therefore truly dynamic.
- For n = 3, the policy improves both average terminal wealth and expected utility.
- For n = 4, wealth improvement alone is not enough; utility-based evaluation remains essential.
- The tabular method is transparent but suffers from severe computational scaling.